

# LC 1034: Coloring A Border

Booking.com, Amazon

## Concept & Notes

- It is an advanced version of color fill.
- In color fill we had to cover all pieces which were connected.
- Here we need to consider **only bordering cells**.

This is same as flood fill with one additional question: is the current cell a border?

## Examples

Eg 1:



## Eg 2:



## DFS Logic Flow

```
dfs() {  
    mark visited  
    determine if border ----> [Border Conditions]  
    dfs(neighbour)  
    if(border) {  
        recolor  
    }  
}
```

### Border Conditions:

- $r == 0$
- $c == 0$
- $r == n - 1$
- $c == m - 1$  (or  $n-1$ )
- **OR** adjacent cell is of different colour

## C++ Implementation

---

```

class Solution {
public:

    vector<pair<int,int>> dir={{1,0},{-1,0},{0,1},{0,-1}};
    int n,m;
    int orig;

    vector<pair<int,int>> border; //isme saare border element push krenge

    void dfs(int r,int c,vector<vector<int>>& grid,vector<vector<int>>& seen){

        seen[r][c]=1;
        bool isBorder=false;    //check karenge ki kya ye cell border he

        for(auto it:dir){
            int nr=r+it.first;
            int nc=c+it.second;

            //touches outside grid
            if(nr<0 || nr>=n || nc<0 || nc>=m){
                isBorder=true;
                break;
            }

            //different colour neighbour
            if(grid[nr][nc]!=orig){
                isBorder=true;
                break;
            }
        }

        if(isBorder){
            border.push_back({r,c});
        }

        //continue DFS
        for(auto it:dir){

            int nr=r+it.first;
            int nc=c+it.second;

            if(nr>=0 && nr<n && nc>=0 && nc<m){

```

```

        if(grid[nr][nc]==orig && !seen[nr][nc]){
            dfs(nr,nc,grid,seen);
        }
    }
}

vector<vector<int>> colorBorder(vector<vector<int>>& grid, int row, int col, int color) {

    n=grid.size();
    m=grid[0].size();

    orig=grid[row][col];

    vector<vector<int>> seen(n,vector<int>(m,0));

    dfs(row,col,grid,seen); //ham dfs sirf aur sirf border cells mark krne ke liye

    for(auto cell:border){
        grid[cell.first][cell.second]=color;
    }
    return grid;
}
};

```